

Ricardo Andrés Calvo Méndez

Ph.D. Student in Electrical and Computer Engineering
Elmore Family School of Electrical and Computer Engineering
Purdue University
West Lafayette, IN 47907

+1 (765) 595-1110
✉ rcalvome@purdue.edu
🌐 rcalvom.github.io
📄 ricardo-calvo
🔗 rcalvom

SUMMARY

Ph.D. student in Electrical and Computer Engineering at Purdue University working on **software security**, **formal verification**, and **program analysis**, with publications at **ACM CCS**, **FSE**, **ASE**, and **MSR**. My research automates memory-safety verification and vulnerability discovery for low-level and embedded software. I bring **3+ years of industry experience** in full-stack development and quality assurance, including leading development and testing teams.

Research interests: formal verification; program analysis and testing; performance optimization.

EDUCATION

Ph.D. in Electrical and Computer Engineering <i>Purdue University, West Lafayette, IN</i>	Aug 2025 – Expected May 2030
B.Sc. in Systems and Computer Engineering <i>Universidad Nacional de Colombia (ranked #1 in Colombia), Bogotá, Colombia. Class rank: 2/49.</i>	Feb 2018 – Oct 2023

RESEARCH EXPERIENCE

Graduate Research Assistant <i>Purdue University — Advisor: James C. Davis</i> - Co-develop <i>AutoSOUP</i>, a system that combines large language models with the CBMC bounded model checker to automate memory-safety verification of C components (ACM CCS 2026); contributed to system design. - Designed AutoSOUP's debugging agent, which reads verification errors, diagnoses why a proof fails, and proposes a fix. - Built the experimental setup used to run AutoSOUP's large-scale evaluation on a computing cluster. - Across 177 components from four embedded operating systems, AutoSOUP exposed 66.7% of 60 known CVEs (vs. 28.3% for a frontier coding agent) and uncovered 20 new vulnerabilities reported to maintainers. - Co-authored studies on false-positive crashes in Google's OSS-Fuzz-Gen (FSE 2026) and on how AI agents optimize code (MSR 2026); presenting author of both.	Aug 2025 – Present
Undergraduate Visiting Scholar (UREP-C) <i>Purdue University — Advisor: James C. Davis</i> - With mentorship, designed and implemented the fuzzing strategy of <i>EmNetTest</i>, a packet-validation testing framework, and its evaluation on 4 embedded network stacks, using C, C++ and Python. - Contributed to the manual triage of results; EmNetTest reproduced 12 known vulnerabilities and found 7 new defects (6 CVEs), leading to an ASE 2023 publication.	Feb 2023 – Jul 2023

PROFESSIONAL EXPERIENCE

Test Engineer & QA Analyst <i>Kravata S.A.S., Bogotá, Colombia</i> - Joined as the company's first and only QA engineer and defined its testing strategy, standards, and processes from scratch. - Led testing for the company's 4 most critical services. - Automated unit, integration and end-to-end tests with Python, Cucumber, Selenium and Flask, and integrated them into CI/CD pipelines.	Sep 2024 – Jul 2025
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Full-Stack Software Developer

Aug 2023 – Aug 2024

Fit Ideas S.A.S., Bogotá, Colombia

- Promoted to team lead; led **4 developers** building and deploying a **white-label** cross-platform fuel sales application, customizable per client brand.
- Defined the system architecture, ran code reviews, and supported the team's Scrum process.
- Built web and mobile systems with Django, React, Spring Boot, Flutter and PostgreSQL, including end-to-end tests.

Junior Software Developer

Aug 2021 – Jan 2023

Fit Ideas S.A.S., Bogotá, Colombia

- Built a configurable web scraper that collects public online data on demand.
- Developed web and mobile product features with Django, React, Spring Boot and Flutter, and designed and maintained PostgreSQL databases.

TEACHING EXPERIENCE

Graduate Teaching Assistant

Fall 2026

ECE 30861: Software Engineering — Purdue University

- Grade assignments and hold office hours for a course of ~150 students.

TECHNICAL SKILLS

Research & Security: Formal verification, memory-safety verification, bounded model checking (CBMC), program analysis, fuzzing, performance optimization

Programming: C, C++, Python, Java, JavaScript, TypeScript, Dart, SQL

AI for SE: LLM agents, agentic pipelines, empirical evaluation

Engineering: Spring Boot, React, Django, Flask, Flutter, PostgreSQL, Selenium, Cucumber, CI/CD, Docker, Linux, Git

REFEREED CONFERENCE PUBLICATIONS

- [C-1] P.C. Amusuo, R. Calvo, D. Anandayuvraj, T. Le Lievre, K. Kolyakov, E. Jorgensen, A. Machiry, and J.C. Davis. *AutoSOUP: Safety-Oriented Unit Proof Generation for Component-level Memory-Safety Verification* *arXiv*. Accepted at the 33rd ACM Conference on Computer and Communications Security (CCS'26). *Presenting author*.
- [C-2] P. Amusuo, D. Liu, R. Calvo, J. Metzman, O. Chang, and J.C. Davis. *Lessons from Mitigating False Positives in Google's OSS-Fuzz-Gen* *DOI*. Proceedings of the ACM International Conference on the Foundations of Software Engineering (FSE) (FSE-Industry'26). *Presenting author*.
- [C-3] H. Peng, A. Z. Qiu, R. Calvo, K.G. Kalu, and J.C. Davis. *How Do Agents Perform Code Optimization? An Empirical Study* *DOI*. Proceedings of the 23rd International Conference on Mining Software Repositories — Mining Challenge track (MSR-Mining'26). *Presenting author*.
- [C-4] P.C. Amusuo, R. Calvo, Z. Xu, A. Machiry, and J.C. Davis. *Systematically Detecting Packet Validation Vulnerabilities in Embedded Network Stacks* *DOI*. Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering (ASE'23).

PRELIMINARY WORKS

- [P-1] P.C. Amusuo, D. Liu, R. Calvo, J. Metzman, O. Chang, and J.C. Davis. *FalseCrashReducer: Mitigating False Positive Crashes in OSS-Fuzz-Gen Using Agentic AI* *arXiv*. Technical preprint (2025).

ACADEMIC SERVICE

Reviewer — MSR'26 Mining Challenge Track	2026
Sub-reviewer — ICSE'26, ICSE'27, USENIX Security '26 (Cycle 2)	2026
Student Volunteer — ICSE'26	2026

GRANTS AND SCHOLARSHIPS

CCS'26 Student Conference Grant <i>ACM CCS</i> <i>Travel support for CCS in The Hague, Netherlands.</i>	2026
ICSE'26 Student Mentoring Workshop (SMeW) <i>National Science Foundation (NSF) and ACM SIGSOFT</i> <i>Travel support for the Student Mentoring Workshop in Rio de Janeiro, Brazil.</i>	2026
“Ser Pilo Paga” Scholarship <i>Ministry of Education of Colombia</i> <i>Merit scholarship for a 99th-percentile national exam score; full undergraduate tuition.</i>	2017